LIFTING AN INVERSE SYSTEM OF OBJECTS OF D(AB) TO A SYSTEM OF COMPLEXES WHOSE TRANSITION DIAGRAMS COMMUTE STRICTLY

THE STACKS PROJECT AUTHORS

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Editorial note. Both the statement and the proof display a diagram in the derived category with Xy-pic, using the author's own markup and the declaration of Ab and of N from the retained preamble; nothing about the diagrams was changed.

Editorial note. The lead-in text at source lines 23330-23332, contrasting an inverse system of complexes with an inverse system of objects of D(Ab) and announcing that the next lemma and remark explain the difference, is outside the counted spans because it belongs to the surrounding discussion.

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Allowed standard prerequisites (editorial).

- The objects of D(Ab) are complexes of abelian groups up to homotopy of maps, and a morphism in D(Ab) between two complexes is represented by a morphism of complexes after resolving the source by a complex of projectives. used at proof lines 23363-23367, where the author appeals to the definition of morphisms in D(Ab) to find a complex and a morphism of complexes representing a prescribed composition. Supplies the transition map of complexes that represents the composite in D(Ab), which is what makes the resulting system a system of complexes rather than a system of derived objects.
- The category D(Ab(N)) of inverse systems of complexes of abelian groups has as morphisms the levelwise morphisms of complexes, and two such systems are isomorphic when every level is an isomorphism in D(Ab) and

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the transition maps agree in D(Ab). used at proof lines 23336-23347, where the object M = (M_n) of D(Ab(N)) and the isomorphisms M_n -> K_n are required. Fixes the exact form of the lifting that has to be constructed level by level.

- An inverse system indexed by the natural numbers may be constructed by induction on the length of its finite initial segments. used at proof lines 23351-23355 and closed at proof line 23372. Turns the levelwise construction into an induction, with the case of length one as the initial step.

Lemma 0.2. *Let (K_n) be an inverse system of objects of $D(Ab)$. Then there exists an object $M = (M_n^\bullet)$ of $D(Ab(\mathbf{N}))$ and isomorphisms $M_n^\bullet \to K_n$ in $D(Ab)$ such that the diagrams*

$$\begin{array}{ccc} M_{n+1}^\bullet & \longrightarrow & M_n^\bullet \\ \downarrow & & \downarrow \\ K_{n+1} & \longrightarrow & K_n \end{array}$$

commute in $D(Ab)$.

Proof. Namely, let $M_1^\bullet$ be a complex of abelian groups representing K_1. Suppose we have constructed $M_e^\bullet \to M_{e-1}^\bullet \to \ldots \to M_1^\bullet$ and maps $\psi_i : M_i^\bullet \to K_i$ such that the diagrams in the statement of the lemma commute for all $n < e$. Then we consider the diagram

$$\begin{array}{ccc} & & M_n^\bullet \\ & & \downarrow{\scriptstyle \psi_n} \\ K_{n+1} & \longrightarrow & K_n \end{array}$$

in $D(Ab)$. By the definition of morphisms in $D(Ab)$ we can find a complex $M_{n+1}^\bullet$ of abelian groups, an isomorphism $M_{n+1}^\bullet \to K_{n+1}$ in $D(Ab)$, and a morphism of complexes $M_{n+1}^\bullet \to M_n^\bullet$ representing the composition

$$K_{n+1} \to K_n \xrightarrow{\psi_n^{-1}} M_n^\bullet$$

in $D(Ab)$. Thus the lemma holds by induction. □